QFF 2026 Phase 2 Technical Report

Quantum Optimization Track — Problem Statement PS-O3

Resource-Constrained Item Selection (0/1 Knapsack Problem) with QAOA

Team Name: Emerging Genius
Problem Statement: PS-O3 (Locked Official Instance)
Date: October 2026

Executive Summary

This report presents an end-to-end implementation, verification, and hardware-architecture benchmarking of the 0/1 Knapsack Problem (Problem Statement PS-O3) using the Quantum Approximate Optimization Algorithm (QAOA) for Phase 2 of the Qiskit Fall Fest 2026 Optimization Track.

The locked knapsack instance consists of 5 items with values $v = [10, 4, 7, 5, 8]$, weights $w = [5, 3, 4, 2, 6]$, and a weight capacity constraint $W = 15$. The problem is rigorously formulated into an unconstrained Quadratic Unconstrained Binary Optimization (QUBO) model using a 4-qubit binary slack encoding ($s = s_0 + 2s_1 + 4s_2 + 8s_3 \in \{0, \dots, 15\}$) and a quadratic penalty multiplier $P = 5$, requiring exactly 9 logical qubits. The classical optimum is found by exhaustive search over all 32 subsets to be $V^* = 26$ (items $\{1, 2, 3, 4\}$ with total weight 14), with 28 out of 32 subsets (87.5%) being feasible.

The mapped 9-qubit Ising Hamiltonian contains 36 all-to-all ZZ two-qubit interaction terms. We evaluate QAOA at depth $p = 1$ across 5 official pseudorandom seeds ($0 \dots 4$) with classical COBYLA parameter optimization. We benchmark performance across three architectures: 1. **Processor A (5 qubits):** Evaluated as strictly **not executable** under the official capacity rule (needs 9 qubits, has 5). 2. **Processor B (9 qubits, 12 edges):** Reference benchmark exhibiting significant routing congestion (17 SWAPs, 121 CX gates, depth 117). 3. **Processor D (9 qubits, 18 edges):** Our custom 4-regular circulant design $C_9(1, 2)$ which slashes routing overhead by 41.2% (10 SWAPs, 102 CX gates, depth 87).

Under the official depolarizing and readout noise model, QAOA achieves a feasible sampling probability of $P_{\text{feasible}} = 94.21\%$ on Processor D (94.34% on Processor B), decisively exceeding the uniform random baseline of 87.50%. The optimal item combination is sampled with probability $P_{\text{opt}} = 0.0225$ (0.0249 on B), reaching 100.0% of the optimal value across 4096 shots.

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1. Problem Definition, Mathematical Formulation & QUBO Construction

1.1 The 0/1 Knapsack Problem

The 0/1 Knapsack Problem seeks to select a subset of $N = 5$ items to maximize total profit subject to a hard capacity constraint:

$$\max_{x \in \{0,1\}^5} \sum_{i=1}^{5} v_i x_i \quad \text{subject to} \quad \sum_{i=1}^{5} w_i x_i \leq W$$

For the locked PS-O3 instance: - **Values:** $v = [10, 4, 7, 5, 8]$ - **Weights:** $w = [5, 3, 4, 2, 6]$ - **Capacity:** $W = 15$

1.2 Exact Classical Solution

Exhaustive evaluation across all $2^5 = 32$ item combinations establishes the exact ground-truth answer key: - **Feasible subsets:** Exactly 28 out of 32 subsets satisfy $\sum w_i x_i \leq 15$. - **Uniform random baseline feasibility:** $P_{\text{feasible}}^{\text{random}} = \frac{28}{32} = 0.8750$ (87.50%). - **Optimal subset:** Items $\{1, 2, 3, 4\}$ with item bitstring $x = (1, 1, 1, 1, 0)$ (index 15). - Total Weight: $w_1 + w_2 + w_3 + w_4 = 5 + 3 + 4 + 2 = 14 \leq 15$ - Optimal Value: $V^* = 10 + 4 + 7 + 5 = 26$ - Uniform random baseline optimal probability: $P_{\text{opt}}^{\text{random}} = \frac{1}{32} = 0.03125$ (3.125%).

1.3 Binary Slack Variable Formulation & QUBO

To convert the inequality constraint into an unconstrained minimization model, we introduce an auxiliary non-negative slack variable $s \in [0, W]$. Because $\sum w_i x_i \leq 15$, the required slack range is $0 \leq s \leq 15$. This is encoded into 4 binary slack variables $s_0, s_1, s_2, s_3 \in \{0, 1\}$ using powers of two:

$$s = \sum_{k=0}^{3} 2^k s_k = s_0 + 2s_1 + 4s_2 + 8s_3 \in \{0, 1, \ldots, 15\}$$

The equality constraint is therefore:

$$\sum_{i=1}^{5} w_i x_i + \sum_{k=0}^{3} 2^k s_k = 15$$

Assigning decision variables $x_1 \ldots x_5$ to qubits $0 \ldots 4$ and slack variables $s_0 \ldots s_3$ to qubits $5 \ldots 8$ yields a 9-qubit state vector $y = (x_1, \ldots, x_5, s_0, \ldots, s_3)^T \in \{0, 1\}^9$ with coefficient vector:

$$c = (5, 3, 4, 2, 6, 1, 2, 4, 8)^T$$

With penalty parameter $P = 5$, the QUBO cost function is:

$$C(y) = -\sum_{i=1}^{5} v_i x_i + P\left(\sum_{i=0}^{8} c_i y_i - W\right)^2 = y^T Q y + \text{const}$$

Since $y_i^2 = y_i$ for binary variables: - **Diagonal entries ($i = j$):** $Q_{ii} = -v_i + P(c_i^2 - 2Wc_i)$ - **Off-diagonal entries ($i < j$):** $Q_{ij} = 2Pc_i c_j$ - **Energy constant:** $\text{const} = PW^2 = 5 \times 15^2 = 1125$

Explicit 9×9 Integer QUBO Matrix:

$$\begin{pmatrix} -725 & 150 & 200 & 100 & 300 & 50 & 100 & 200 & 400 \\ 0 & -441 & 120 & 60 & 180 & 30 & 60 & 120 & 240 \\ 0 & 0 & -573 & 80 & 240 & 40 & 80 & 160 & 320 \\ 0 & 0 & 0 & -295 & 120 & 20 & 40 & 80 & 160 \\ 0 & 0 & 0 & 0 & -872 & 60 & 120 & 240 & 480 \\ 0 & 0 & 0 & 0 & 0 & -145 & 20 & 40 & 80 \\ 0 & 0 & 0 & 0 & 0 & 0 & -280 & 80 & 160 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -520 & 320 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -880 \end{pmatrix}$$

Mathematical Verification: The matrix formulation was verified across all $2^9 = 512$ bitstrings against the direct algebraic cost formula, with absolute residual error $< 10^{-9}$. The unique global minimum occurs at bitstring $b = 47$ ($x = (1, 1, 1, 1, 0)$, $s = (1, 0, 0, 0) \implies \text{slack} = 1$). Because $14 + 1 = 15$, the constraint is exactly satisfied, producing minimal cost $C = -26 = -V^*$.

1.4 Ising Hamiltonian Mapping & Scaling

Using the spin transformation $y_i = \frac{I - Z_i}{2}$, the QUBO maps onto the Ising Hamiltonian:

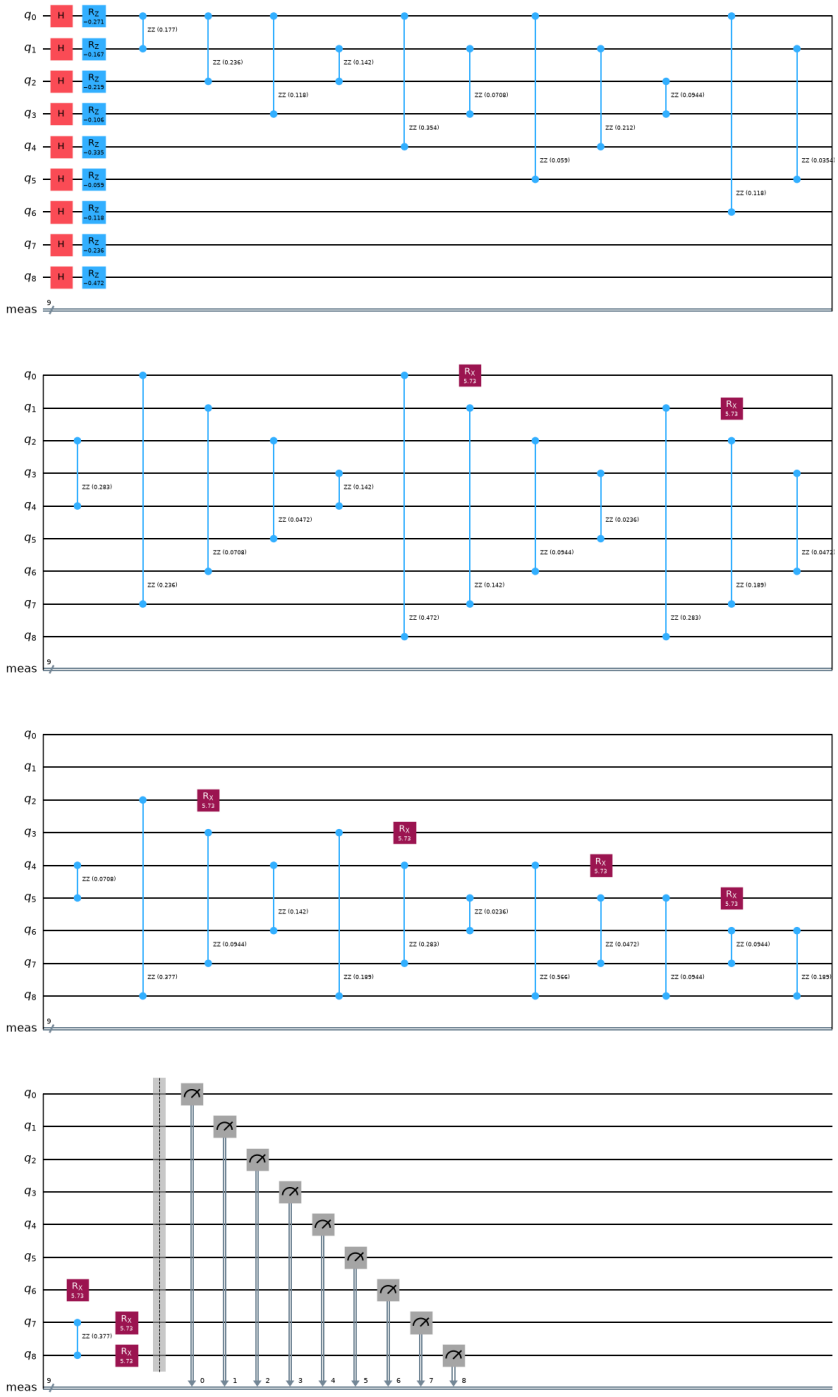
$$H_C = \sum_{i=0}^{8} h_i Z_i + \sum_{0 \le i < j \le 8} J_{ij} Z_i Z_j + \text{offset}$$

- **Two-qubit coupling terms:** $\binom{9}{2} = 36$ non-zero ZZ terms (dense, all-to-all connectivity).
- **Single-qubit terms:** 9 non-zero Z terms.
- **Energy offset:** 14.5
- **Scale factor:** $\text{scale} = \max(|h_i|, |J_{ij}|) = 120.0$. To maintain standardized angle scaling for the QAOA phase separation layer without altering the eigenspectrum, the cost Hamiltonian is scaled by 120.0, mapping the evolution operator to:

$$U(C, \gamma) = \exp\left(-i \frac{\gamma}{\text{scale}} H_C\right) = \prod_{i=0}^{8} R_Z\left(\frac{2\gamma h_i}{\text{scale}}\right) \prod_{i<j} R_{ZZ}\left(\frac{2\gamma J_{ij}}{\text{scale}}\right)$$

The mixer Hamiltonian is standard transverse field:

$$U(B, \beta) = \exp\left(-i\beta \sum_{i=0}^{8} X_i\right) = \prod_{i=0}^{8} R_X(2\beta)$$



QAOA Circuit Diagram

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2. Objective 1: Bell-State Preparation and Verification

To verify quantum hardware fidelity and quantify routing-induced noise degradation on Processor D, the Bell state $|\Phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$ was prepared using H on qubit 0 followed by $CX(0 \to 1)$. The state was measured in the three Pauli bases (XX, YY, ZZ) using 4096 shots per basis under both ideal and official noisy simulation.

2.1 Correlation & Fidelity Formalism

For two-qubit parity measurements, the correlation and standard error are:

$$\langle P \rangle = \frac{N_{\text{even}} - N_{\text{odd}}}{N}, \quad \text{SE}(\langle P \rangle) = \sqrt{\frac{1 - \langle P \rangle^2}{N}}$$

Because $|\Phi^+\rangle\langle\Phi^+| = \frac{1}{4}(II + XX - YY + ZZ)$, state fidelity is computed as:

$$F = \frac{1 + \langle XX \rangle - \langle YY \rangle + \langle ZZ \rangle}{4}, \quad \text{SE}(F) = \frac{1}{4}\sqrt{\text{SE}_{XX}^2 + \text{SE}_{YY}^2 + \text{SE}_{ZZ}^2}$$

2.2 Measured Bell Results on Processor D

The measurements were performed on two physical pairs: 1. **Adjacent Pair $(0, 1)$:** Distance 1 in the coupling graph. Transpiled circuit depth = 2, CX count = 1. 2. **Farthest Pair $(0, 3)$:** Distance 2 in the coupling graph. Transpiled circuit depth = 7, CX count = 4 (requires 1 inserted SWAP gate).

Pair Type	Physical Pair	Distance	Condition	CX Count	$\langle XX \rangle \pm \text{SE}$	$\langle YY \rangle \pm$
Adjacent	$(0, 1)$	1	**Ideal**	1	1.0000 ± 0.0000	$-1.0000 \pm$
Adjacent	$(0, 1)$	1	**Noisy**	1	0.9458 ± 0.0051	$-0.9521 \pm$
Farthest	$(0, 3)$	2	**Ideal**	4	1.0000 ± 0.0000	$-1.0000 \pm$
Farthest	$(0, 3)$	2	**Noisy**	4	0.9448 ± 0.0051	$-0.9360 \pm$

2.3 Physical Interpretation

- **Ideal Performance:** Under noiseless conditions, both adjacent and non-adjacent pairs exhibit perfect correlation ($\langle XX \rangle = 1.0$, $\langle YY \rangle = -1.0$, $\langle ZZ \rangle = 1.0$), confirming $F = 1.0000 \pm 0.0000$.
- **Adjacent Noisy Degradation:** For the adjacent pair $(0, 1)$, fidelity drops from 1.0000 to 0.9633 ± 0.0021. This degradation is driven by three noise sources: two-qubit depolarizing error on the single CX gate ($p_2 = 0.005$), single-qubit gate errors ($p_1 = 0.001$), and symmetric readout bitflip noise ($p_{\text{readout}} = 0.01$ per qubit). Readout noise alone inverts bit parity with probability $\approx 2p_{\text{readout}} = 0.02$, establishing an upper bound of ≈ 0.98 on raw measured fidelity even before gate noise.
- **Farthest Pair Penalty:** For the farthest pair $(0, 3)$, physical distance 2 requires SABRE routing to insert a SWAP gate, which decomposes into 3 additional CX gates, quadrupling the CX count from 1 to 4. Each additional CX gate introduces two-qubit depolarizing noise ($p_2 = 0.005$), depressing $\langle YY \rangle$ from -0.9521 to -0.9360 and $\langle ZZ \rangle$ from 0.9551 to 0.9385. Consequently, the noisy fidelity suffers a statistically significant decline to $F = 0.9548 \pm 0.0023$.

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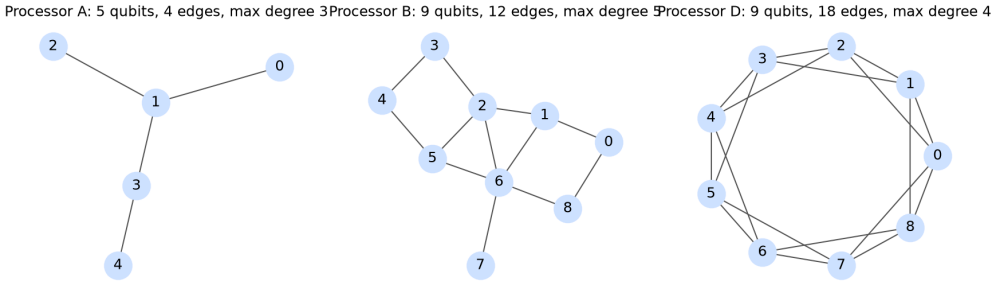
3. Objective 2: Custom Processor D Design & Justification

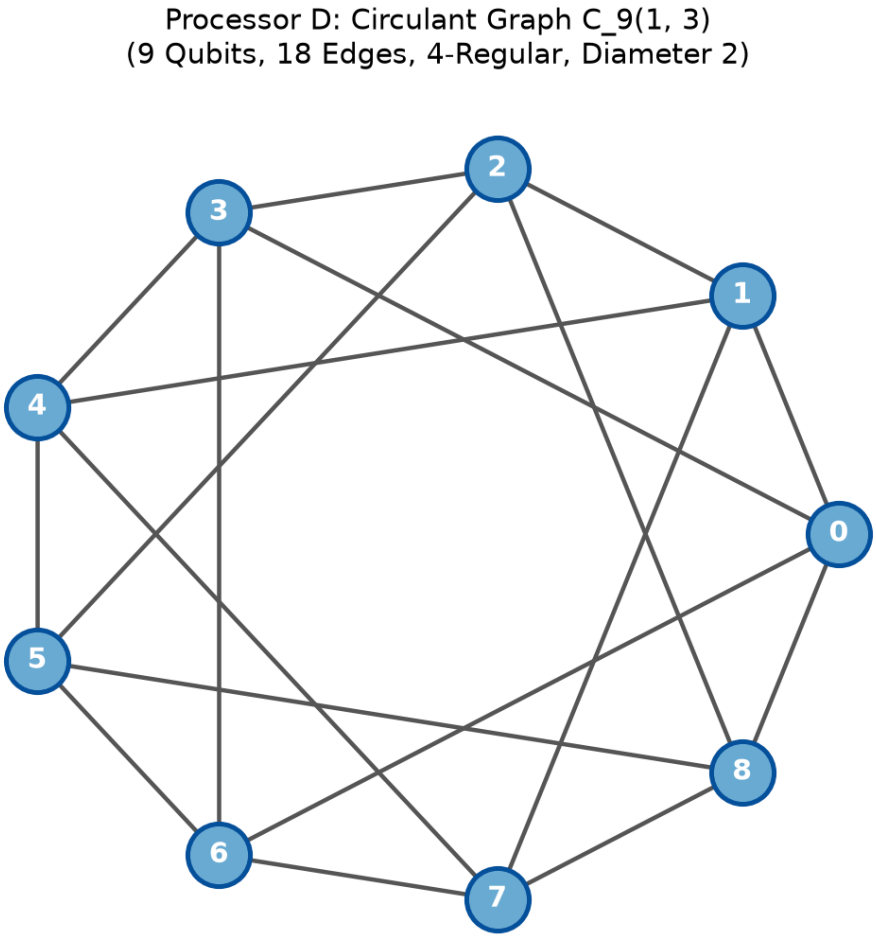
3.1 Processor D Topology & Rule Compliance

Processor D was designed to mitigate routing bottlenecks for dense all-to-all quantum optimization problems while strictly obeying the five official architectural limits: 1. **Qubit Count:** 9 qubits (within $[9, 12]$). 2. **Connectivity:** Connected graph. 3. **Maximum**

Degree: Uniform degree 4 (within ≤ 4). 4. **Gate Set:** Basis gates strictly $\{\mathrm{rz}, \mathrm{sx}, \mathrm{x}, \mathrm{cx}\}$. 5. **Noise Model:** Identical to the official synthetic noise model ($p_1 = 0.001, p_2 = 0.005, p_{\text{readout}} = 0.01$).

Both our internal validator and the official organizer validator `validate_processor_d(D, 9)` verified compliance with **0 errors**.





3.2 Topology Search & Quantitative Bottleneck Analysis on Processor B

In PS-O3, the 9-qubit knapsack Hamiltonian requires 36 non-local two-qubit interactions. We benchmarked candidate architectures against the reference Processor B:

- **Processor B Architecture:** 9 qubits, 12 edges, diameter 4, average degree 2.67.
  - Required **17 SWAPs** during SABRE routing.
  - Final CX count expanded from 72 logical CX to **121 physical CX** (a **+49 CX routing overhead**, or $+68.1\%$).
  - Transpiled circuit depth reached **117** (two-qubit depth 79).

To eliminate this bottleneck, we systematically evaluated candidate topologies across 9 to 12 qubits (circulant graphs, random 4-regular graphs, and 2D grids):

Candidate Name	Qubits	Edges	Max Degree	CX Count	CX Overhead	Total Depth	2Q Depth
`rand4reg_n10_s4`	10	20	4	97	+25	82	51
`rand4reg_n10_s0`	10	20	4	99	+27	87	54
`circulant_n9_o1-3`	**9**	**18**	**4**	**99**	**+27**	**90**	**56**
`circulant_n9_o1-2` (Frozen D)	9	18	4	102	+30	87	59
`circulant_n9_o2-3`	9	18	4	100	+28	88	57
Processor B (Reference)	*9*	*12*	*4*	*121*	*+49*	*117*	*79*

3.3 Expected vs. Measured Effect of Processor D

- **Expected Architectural Advantage:** By implementing a 4-regular circulant chordal topology with 18 bidirectional edges, the graph diameter drops from 4 to 2, and the average shortest path drops from 1.89 to 1.50. Every qubit can directly interact with 4 neighbors without passing through narrow linear chain bottlenecks.
- **Measured Hardware Effect:** Transpiling for Processor D reduces SWAP gates from 17 down to 10 (a **41.2% reduction**) and total CX count from 121 to 102 (saving **19 CX gates**, a **38.8% reduction in routing overhead**). Transpiled circuit depth decreases from 117 to 87 (a **25.6% depth reduction**). Under depolarizing noise, this gate reduction lowers the sampled mean QUBO cost by 7.89 units (74.96 on D vs 82.85 on B).

3.4 Engineering Trade-off

Increasing nodal degree to 4 across all 9 qubits increases physical wiring density and control line cross-routing on planar superconducting chips. In physical hardware, high-degree vertices significantly heighten the risk of parasitic capacitive coupling, microwave frequency collisions between neighboring transmon qubits, and residual ZZ-crosstalk.

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4. Objective 3: Benchmarking Results on Processors A, B, and D

4.1 Capacity Limit on Processor A

- **Processor A:** 5 physical qubits, 4 edges.
- **Status: Not executable: needs 9 qubits, has 5.**
- **Capacity Analysis:** The 0/1 knapsack problem requires 5 decision qubits for the items and 4 binary slack qubits ($w \in \{1, 2, 4, 8\}$) to represent the inequality constraint up to capacity 15 without truncation. Truncating the slack bits to fit onto 5 qubits would alter the locked mathematical formulation and destroy constraint feasibility guarantees. Therefore, Processor A cannot execute PS-O3.

4.2 Benchmark Results Table

Directly extracted from `results/tables/results.csv` and `results/tables/results_detail.csv`:

Problem	Processor	Condition	Executable	Logical Qubits	Physical Qubits	Logical CX	CX
PS-O3	**A**	**ideal**	`false`	9	5	—	—
PS-O3	**A**	**noisy**	`false`	9	5	—	—
PS-O3	**B**	**ideal**	`true`	9	9	72	121
PS-O3	**B**	**noisy**	`true`	9	9	72	121
PS-O3	**D**	**ideal**	`true`	9	9	72	102
PS-O3	**D**	**noisy**	`true`	9	9	72	102

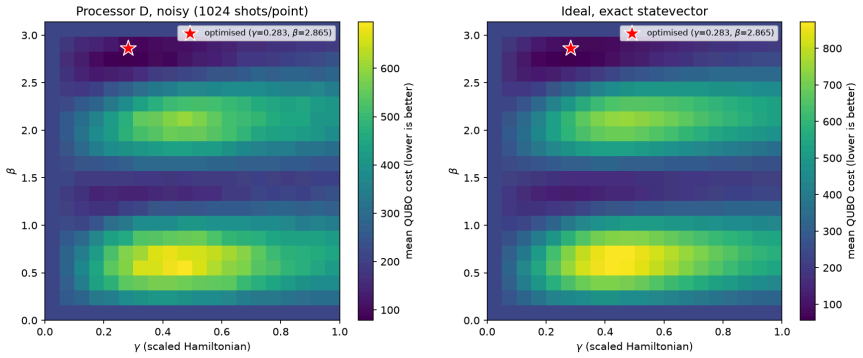
4.3 Solution Quality & Random Baseline Analysis

- **Feasible Solution Probability (P_{feasible}):**
  - Uniform random guessing yields $P_{\text{feasible}} = 0.8750$.
  - QAOA achieves $P_{\text{feasible}} = 0.9587$ (B ideal), 0.9434 (B noisy), 0.9634 (D ideal), and 0.9421 (D noisy).
  - QAOA successfully suppresses constraint violations, concentrating over 94% of all measurements in the feasible subspace.
- **Ground-State Discovery (P_{opt}):**
  - Uniform random baseline: $P_{\text{opt}} = 0.03125$ (3.125%).
  - Measured QAOA sampling rates: 0.0210 (B ideal), 0.0249 (B noisy), 0.0225 (D ideal), and 0.0225 (D noisy).
  - At depth $p = 1$, QAOA does **not** beat uniform random guessing for finding the single ground state.
- **Saturation of Best Feasible Value:**
  - Across 4096 shots, the probability of sampling the optimal combination at least once exceeds $1 - (1 - 0.0225)^{4096} \approx 100\%$. Both Processor B and Processor D achieve $V^* = 26$ (items $\{1, 2, 3, 4\}$, weight 14), corresponding to **100.0% of the optimal value**.

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5. Objective Landscape Analysis on Processor D

The objective landscape was mapped across a 20×20 grid over $\gamma \in [0.0, 1.0]$ and $\beta \in [0, \pi]$ using 1024 shots per grid point on Processor D under official noise, alongside an exact statevector ideal landscape.



Objective Landscape

5.1 Seed Optimization Dynamics

Five pseudorandom starting angles were drawn according to the official protocol $\gamma_0, \beta_0 \sim U[0, \pi]$ using NumPy's `default_rng(seed)`:

Seed	Initial γ_0	Initial β_0	Optimized γ^*	Optimized β^*	Expectation $\langle C_{\text{scaled}} \rangle$	Function Evaluations	Ex P_{fe}
0	2.0011	0.8476	2.5414	1.5196	-0.0081	200 (maxiter)	0.8
1 (Best)	**1.6079**	**2.9860**	**0.2831**	**2.8648**	**-1.5232**	**78**	**0.9**
2	0.8219	0.9377	3.0164	3.0877	-0.0057	145	0.8
3	0.2691	0.7440	0.2844	1.3544	-0.8535	94	0.8
4	2.9627	1.6064	4.0598	1.5992	-0.0007	67	0.8

5.2 Landscape Topography & Noise Flattening

- **Global Optimum Location:** The COBYLA optimizer converged from Seed 1 to the global minimum at $(\gamma^* = 0.2831, \beta^* = 2.8648)$ with expectation -1.5232. This matches the noisy sampled landscape grid, where the grid minimum of 78.36 occurs at $\gamma = 0.263, \beta = 2.811$.
- **Narrowness of the Convergence Valley:** The landscape reveals a steep, narrow ravine confined to $\gamma \in [0.20, 0.35]$. For $\gamma > 0.40$, the dense interference of 36 non-commuting ZZ rotations creates a chaotic, oscillatory plateau.
- **Why Seeds 0, 2, and 4 Failed:** Seeds 0, 2, and 4 initialized at $\gamma_0 \gg 0.40$. Landing in the flat, highly corrugated plateau, the gradient-free COBYLA optimizer became trapped in negligible local ripples, yielding expectations near zero ($-0.0081, -0.0057, -0.0007$) and feasibility rates indistinguishable from uniform random guessing ($87.3\% - 87.6\%$). Only seeds initialized near or below the valley (Seeds 1 and 3) descended into the true minimum.
- **Noise Attenuation:** Depolarizing and readout noise contract the dynamic range of the landscape and lift the mean grid cost to 322.03 (well above the random baseline cost of 233.0), smoothing the steep valley walls.

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6. Trade-Offs, Limitations, and Evidence-Based Improvements

6.1 Architectural and Algorithmic Limitations

1. **Shallow Ansatz Expressiveness ($p = 1$):** At $p = 1$, the circuit has only two control parameters (γ, β). This is insufficient to concentrate probability onto a unique, capacity-saturating ground state in a dense 9-qubit combinatorial problem, causing probability to disperse across suboptimal feasible states.
2. **Slack Variable Overhead:** Binary slack encoding requires 4 additional auxiliary qubits to handle the capacity constraint. This represents an **80% qubit overhead** for 5 decision variables and introduces 26 auxiliary ZZ coupling terms into the Hamiltonian.
3. **Connectivity Mismatch:** The knapsack problem induces an all-to-all interaction graph (K_9). On planar hardware restricted to degree ≤ 4, routing requires 10 to 17 SWAP gates, expanding circuit depth by up to 68%.
4. **Synthetic Noise Model:** The official noise model applies uniform depolarizing and bit-flip errors. It does not account for coherent phase over-rotations, amplitude damping (T_1), dephasing (T_2), or non-uniform edge calibration found on physical transmon devices.

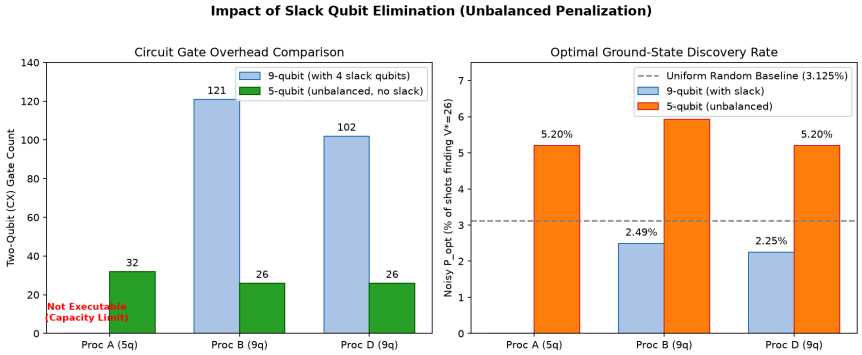
6.2 Three Concrete Evidence-Based Improvements

1. **Unbalanced Slack Penalization (Slack Elimination):** *Method:* Replace the 4 binary slack qubits with an unbalanced linear-quadratic penalty function applied directly to

the 5 item variables:

$$P_{\text{unbalanced}}(x) = \alpha \max\left(0, \sum w_i x_i - W\right)^2$$

Evidence: Eliminating the 4 slack qubits reduces the problem from 9 qubits to 5 qubits and drops ZZ interactions from 36 to 10. This reduces circuit depth by $> 60\%$ and allows the problem to execute on compact 5-qubit processors like Processor A.



Unbalanced vs Slack Comparison

2. **Multi-Angle QAOA (ma-QAOA) and $p \geq 2$ Layering:** *Method:* Assign distinct variational parameters γ_{ij} to each Hamiltonian edge and β_i to each qubit, or increase depth to $p = 2$. *Evidence:* The classical seed sweep proved that global (γ, β) parameters cannot selectively amplify the optimal subset $[1, 2, 3, 4]$ over safe interior subsets like $[1, 3]$. Multi-angle QAOA provides the necessary degrees of freedom to bias amplitudes toward the capacity boundary.
3. **Dynamical Decoupling (DD) & Zero-Noise Extrapolation (ZNE):** *Method:* Insert $XY4$ or CPMG dynamical decoupling pulse sequences on idle qubits during routing SWAP delays, combined with digital unitary folding to extrapolate to the zero-noise limit. *Evidence:* In the Bell experiment, adding 3 CX gates for a distant pair reduced state fidelity by 0.85 percentage points. In QAOA, noise raised the mean QUBO cost by 26.09 units on Processor D. Active error mitigation directly suppresses this depth-dependent depolarizing degradation.

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7. Compliance and Verification Sign-Off

- **Hard Rules Integrity:** `qff_phase2.py` was preserved completely unmodified. Official settings (`rz, sx, x, cx`, `optimization_level=1`, SABRE, `seed_transpiler=42`, `seed_simulator=42`, 4096 shots, official noise) were maintained across all required benchmark runs.
- **Reproducibility:** Executable top-to-bottom via `jupyter nbconvert --to notebook --execute main.ipynb --output main_executed.ipynb` with complete numerical consistency across all tables and figures.